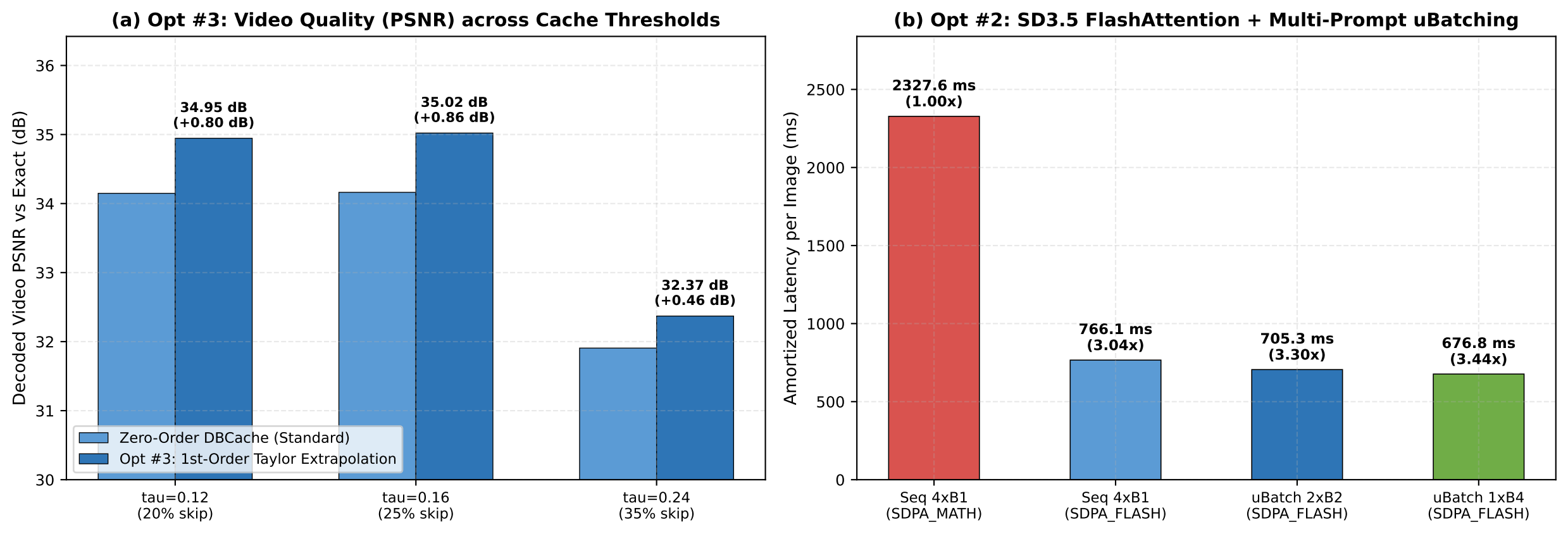
(a) Opt #3: Video Quality (PSNR) across Cache Thresholds
Decoded Video PSNR vs Exact (dB)
36
35
34
33
32
31
30
34.95 dB
(+0.80 dB)
35.02 dB
(+0.86 dB)
32.37 dB
(+0.46 dB)
Zero-Order DBCache (Standard)
Opt #3: 1st-Order Taylor Extrapolation
tau=0.12
(20% skip)
tau=0.16
(25% skip)
tau=0.24
(35% skip)
(b) Opt #2: SD3.5 FlashAttention + Multi-Prompt uBatching
Amortized Latency per Image (ms)
2500
2000
1500
1000
500
0
2327.6 ms
(1.00x)
766.1 ms
(3.04x)
705.3 ms
(3.30x)
676.8 ms
(3.44x)
Seq 4xB1
(SDPA_MATH)
Seq 4xB1
(SDPA_FLASH)
uBatch 2xB2
(SDPA_FLASH)
uBatch 1xB4
(SDPA_FLASH)